



Assignment Brief (RQF)

Higher National Diploma in Computing

Unit 7: Software Development Lifecycles

Assignment Brief

Student Name/ID Number	
Unit Number and Title	Unit 7: Software Development Lifecycles
Academic Year	2026
Unit Tutor	Ms. Yee Mon
Assignment Title	Assignment 1 of 2: Software Development Lifecycles and Feasibility Study Analysis
Issue Date	27/4/2026 (Monday)
Submission Date	24/5/2026 (Sunday)
Submission Format	
The submission is in the form of an individual written report and presentation. You should submit the report in pdf and presentation file in pdf. This should be written in a concise, formal business style using single spacing and font size 12. You are required to make use of headings, paragraphs and subsections as appropriate, and all work must be supported with research and referenced using the Harvard referencing	

system. Please also provide a bibliography using the Harvard referencing system.

- Use Microsoft Word for documentation and report.
- Your report must be formatted as follows:

➤	Font Name	:	Verdana
➤	Font Size	:	12 pt
➤	Margin	:	.75 inches (round)
➤	Line Spacing	:	1.5 pt
➤	Alignment	:	Justify
➤	Left Hand Side of Header	:	Batch Number
➤	Right Hand Side of Header	:	Your Name
➤	Centre of Footer	:	Page Number
- Use the Harvard referencing system.
- Complete the title page and sign the statement of authenticity.
- Your assignment must be submitted in pdf format.

Unit Learning Outcomes

The purpose of this assignment is to:

LO1 Describe different software development lifecycles

LO2 Explain the importance of a feasibility study

Transferable skills and competencies developed

On successful completion of this unit, students understand software development lifecycles so that they can demonstrate their knowledge by implementing a software development lifecycle with a suitable methodology. They will examine various lifecycle models and learn to appreciate their characteristics to understand which project environments are most appropriate.

Vocational scenario

You are a junior software development team working at a local IT company in Myanmar.

A university is facing transportation problems. Students complain that campus shuttle buses are:

- ❖ Often late
- ❖ Overcrowded
- ❖ Difficult to track

The university management wants to develop a computerized **Smart Campus Shuttle System**. You are a junior trainee software developer in the IT company. To develop a "**Campus Shuttle System**" for private university, your project manager will tell you, as a team member, that:

1. Which software development lifecycle (SDLC) model should be used
2. Whether a feasibility study is necessary before starting development.

The university management expects to implement a computerized Campus Shuttle System that can fulfil the following activities such as

- ❖ Track buses in real-time
- ❖ Show arrival times via a mobile app
- ❖ Allow students to report issues

The university hires a development team to propose how the system should be developed.

The university gives the IT company 6 months to develop the system.

Assignment activity and guidance

As part of the project team, you've been tasked with writing a project briefing report for your manager that includes the following facts such as SDLC, Risk Management in software models and Feasibility Report Discussion—

Task-1: Software Development Lifecycle Models:

- Describe at least two iterative and two sequential software development lifecycles SDLC models (e.g., Waterfall, Agile, Spiral, V & V Model, etc). You must clarify key characteristics and explain how each model could be applied to custom **Campus Shuttle System** project.
- Explain how risk is managed in software lifecycle models. Use examples relevant to the **Campus Shuttle System** domain and propose strategies for mitigating potential risks.

Task-2: Feasibility Study

- Explain what a feasibility study is and why it is a critical early step in the SDLC of custom **Campus Shuttle System**. Outline key components and their importance in making informed decisions.
- Identify and briefly describe at least four types of feasibility (technical, economic, legal, operational, schedule). Describe how technical solutions can be compared during the feasibility study for **Campus Shuttle System** project depending on the factors such as tracking bus route, reporting bus track, reporting bus issue, etc.
- Propose questions or investigations that would be part of a feasibility study for the **Campus Shuttle System** project.

Moreover, when you are explaining in feasibility studies, you can discuss the components of a feasibility report for **Campus Shuttle System** project.

In addition, provide a detailed report assessing the suitability of the Waterfall model for a large-scale **Campus Shuttle System** project. Analyse advantages, disadvantages, and make recommendations based on the unique requirements of **Campus Shuttle System** Solutions.

Furthermore, develop a detailed report assessing the impact of different feasibility criteria on the software investigation for **Campus Shuttle**

System Solutions. Analyse factors such as technical feasibility, economic viability, and environmental sustainability.

In the **presentation** slides, you must write what software development lifecycle (SDLC) model is chosen for your above scenario and feasibility studies of your **key findings**. All students must present face to face presentation and that presentation no more than **10 minutes** for your **summarized observations**.

Submission Guidelines:

You need to submit **two** files:

1. Presentation slides in a pdf format compatible with common presentation tools.
2. Pdf file format Report: Submit written reports in a standard document format with proper formatting and citation.
3. This assignment is designed to assess your ability to apply software development concepts in real-world scenario. Tailor your recommendations to the unique requirements and objectives of **Campus Shuttle System** Project.
4. The **maximum** word count for this report must be **3000** or **4000**.

Note to Students:

- ☐ Check carefully the submission date and the instructions given with the assignments
- ☐ Late assignments will not be accepted, except for valid circumstances with supporting evidence.
- ☐ Ensure that you give yourself enough time to complete the assignment by the due date.
- ☐ Do not leave things such as printing to the late minute-excuses of this nature will not be accepted for failure to hand-in the work on time.

- ☐ You must take responsibility for managing your own time effectively.
- ☐ If you are unable to hand in your assignment on time and have valid reasons such as illness, you may apply (in writing) for an extension.
- ☐ Take great care that if you use other people's work or ideas in your assignment, you properly reference them in your text and any bibliography.

Note: **Plagiarism** is an offence. Plagiarized work will be deferred.

Interim Review Session (Formative Assessment)

- ☐ 11/5/2026 (Monday)
- ☐ 12/5/2026 (Tuesday)

Submit the work to moodle.gusto-education.com by

- ☐ 11:55 PM on 24/5/2026 (Sunday)

Sources of information:

You must provide of your references, internet sites or book names. Use Harvard referencing for book references.

Pass	Merit	Distinction
LO1 Describe different software development lifecycles		D1 Assess the merits of applying the Waterfall lifecycle model to a large software development project.
P1 Describe two iterative and two sequential software lifecycle models.	M1 Discuss using an example, why a particular lifecycle model is selected for a development environment.	
P2 Explain how risk is managed in software lifecycle models.		
LO2 Explain the importance of a feasibility study		D2 Assess the impact of different feasibility criteria on a software investigation.
P3 Explain the purpose of a feasibility report.	M2 Discuss the components of a feasibility report.	

P4 Describe how technical solutions can be compared.